Cross-Generational Accommodation of Cantonese Sound Change Features Toward Human and AI Interlocutors

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Abstract: Phonetic accommodation can be viewed as a miniature version of sound change; therefore, investigating accommodation can shed light on the current trajectory of an ongoing sound change. While studies of adults' phonetic accommodation reveal how sound systems are shaped by social contexts, examining children's accommodation provides insight into the latest stage of a sound change. Moreover, in the current era, the emergence of Artificial Intelligence (AI) interlocutors introduces an additional source of sound variation. This raises further questions: Can AI interlocutors introduce new variables into sound change, and do children and adults respond differently to these variants? The current study addresses these questions by comparing adults' and children's accommodation to sound change features of Hong Kong Cantonese (HKC) in a shadowing task using either a human face or an avatar face as the interlocutor. The results showed that adults generally converged more than children, conditioned by different features. Convergence patterns were similar across the two interlocutor types, suggesting that AI interlocutors can elicit accommodation patterns comparable to those produced by human interlocutors and may therefore have similar potential to influence sound change. The difference between children and adults remained consistent across interlocutor types.

Keywords: Phonetic accommodation, children, sound change, AI interlocutors

1 Introduction

Based on Ohala's theory of listeners being the origin of sound change [1], sound change is initiated when listeners fail to correct a non-canonical variant and instead adopt it, incorporating it into their own production, and gradually spreading it. Listeners' sensitivity to such variants and their motivations for not correcting them play crucial roles in this process. In the current era, the emergence of Artificial Intelligence (AI) interlocutors introduces an additional source of sound variation. Will humans also adopt and spread the variants produced by AI interlocutors?

Phonetic accommodation, a phenomenon where one talker adopts the speech characteristics of another talker [3], can be viewed as a miniature version of sound change. It offers the means for sound change to spread through a community [4]. Therefore, investigating phonetic accommodation in human-AI interlocutor interaction (HAI) may inform the tendency of sound change in the current era. Theories draw different predictions for accommodation in HAI, compared to human-human interaction (HHI).

Mediated accounts such as the Communication Accommodation Theory (CAT; [5], [6]) posits that facing interlocutors with different social importance and social needs, speakers will adopt different accommodation strategies in interactions. The accommodation patterns between HHI and HAI depend on how speakers perceive AI interlocutors. In contrast, unmediated accounts such as the Interactive Alignment Model (IAM; [7]) posit that the alignment between interlocutors is a low-level and automatic process, independent of speaker identity [6]. Comparable accommodation pattern is expected between HHI and HAI. Extending to sound change, studies on Cantonese have shown that speakers accommodated differently when shadowing tone merger pairs produced by human talkers than AI talkers [8, 9]. Likewise, in some small sample studies, Cantonese speakers also exhibit greater convergence toward human than AI talkers' sound change variants [10, 11].

In the process of spreading sound change, adults were claimed to be the main force due to wider social network and larger social mobility [12]. Studies of adults' phonetic accommodation reveal how a sound system is shaped by social contexts [e.g., 5, 6]. On the other hand, children have been found to acquire sound change variation at a very early age [15]. Examining children's accommodation provides insight into the most up-to-date progress of a sound change as it becomes embedded in the next generation. Studies of children's interactions with AI agents have demonstrated phonetic convergence in phoneme duration [16]. Moreover, research on child-AI communication suggests that children may transfer communication habits developed through interactions with AI (e.g., lower communicative expectations, more tolerance of communication breakdowns) to their daily human interactions [17]. Together, it is reasonable to infer that children may also acquire sound change variants from AI interlocutors and subsequently use them in human interactions.

To address the above issues, we focus on Hong Kong Cantonese (HKC), a variety of Cantonese within the Chinese language family that has been undergoing various sound changes in the past century. Because these changes are at different stages of progression [18], individual sound change features may elicit varying degrees of accommodation. Moreover, based on our survey, most AI-generated speech adopted the standard and conservative variants of HKC probably because these models were trained on standard speech such as news broadcasts and radio programs. HKC thus provides an ideal opportunity to examine whether listeners would respond differently to sound change variants produced by human and AI interlocutors.

Building on these observations, the current study addresses the following research questions: (1) Do HKC native speakers accommodate when shadowing Cantonese words that reflect ongoing sound change? (2) Does the pattern of accommodation differ between AI and human interlocutors, and do children and adults respond differently to these interlocutors? (3) Does the degree of accommodation vary across different sound change features, and do children and adults exhibit similar or distinct patterns?

2 Method

2.1 Data Collection

Seven HKC sound change features were selected to create monosyllabic stimuli (Table 1), six monosyllabic words each. Two thirds of the words are played to the participants using the less frequent variant nowadays, serving as target stimuli. Based on previous literature [18] and our pilot data, these variants were selected because they were either the citation forms of the word that are used infrequently or emerging variants that had not yet become widespread. The remaining were played using the more frequent variants, serving as fillers. The stimuli were recorded by two phonetically trained female HKC native speakers.

Table 1. Ongoing HKC sound change features under investigation.

Sound Change Feature	Less Frequent Variant (target stimuli)	Examples (→less frequent variant)
initial [n]-[l] merger	[n-]	你 [lei5] → [**n**ei5] ‘you’
initial [ŋ] deletion	Ø	愛 [ŋɔi3] → [ɔi3] ‘love’
initial [k]-[k^{w}] merger	[k^{w}-]	哥 [kɔ1] → [**k**^{w}ɔ1] ‘brother’
initial [k^{w}]-[k] merger	[k-]	國 [k^{w}ɔk3] → [**k**ɔk3] ‘country’
syllabic [m̩]-[ŋ] merger	[ŋ̍]	五 [m̩] → [**ŋ̍**5] ‘five’
[ŋ]-[n] coda merger	[-n]	講[kɔŋ2] → [kɔ**n**2] ‘speak’
[n]-[ŋ] coda merger	[-ŋ]	湯 [t^{h}ɔn1] → [t^{h}ɔ**ŋ**1] ‘soup’

Thirty-two native HKC speakers (16 young adults, mean age = 22.4±3.03; 16 children, mean age = 10.75±1.18) participated in a shadowing task [19], which consists of three blocks: a pre-shadowing block, a human interlocutor shadowing block, and an AI interlocutor shadowing block. The order of the two shadowing blocks was randomized and counterbalanced across participants. The stimuli were presented as traditional Chinese characters on the screen. In the pre-shadowing block (as baseline), participants read the word one by one. During the shadowing blocks, the auditory stimulus was presented 300ms before the corresponding Chinese character appeared. Participants were instructed to read the character after listening to the model speaker's production without any other explicit instruction. The stimuli were accompanied by either an image of a human (human interlocutor block) or an image of an avatar generated using Zoom [20] (AI interlocutor block). Thus, in this study, the human interlocutor refers to the human-faced interlocutor, whereas the AI interlocutor refers to the AI-faced interlocutor.

2.2 Data Processing

Two phonetically trained native HKC raters identified the sound change variants produced by the participants in a binary way. Variants that were perceived the same as our target variants were labeled as 'target variant' while variants that matched our filler variants were labeled as 'filler variant'. Cohen's kappa indicated almost perfect agreements between raters for adults' production (κ = 0.89, 95% CI [0.88, 0.90]), and for children's production (κ = 0.89, 95% CI [0.88, 0.91]). The disagreed tokens were sent to a third HKC rater, and the final label was decided based on majority judgement.

Generalized Linear Mixed Models (GLMMs) were fitted using the 'lme4' package [21] in R to examine the factors predicting shadowing results. Participants' baseline production, shadowing production and model talker's production were coded binarily (target variant = 1, filler variant = 0). Following the linear combination approach [22], participants' Shadowing production (code_s) were fitted as dependent variable. Fixed effects included participants' Baseline production (code_b), the Model talker's productions (code_m), sound change features (Feature), age groups (Age; children vs. adults), and interlocutor types (Interlocutor; human vs. AI). To examine whether accommodation differed between human and AI interlocutors, the model included the interaction between the Model talker's productions and Interlocutor type, as well as the three-way interaction among the Model talker's productions, Age group, and Interlocutor type to test whether interlocutor effects differed between children and adults. The model also included the three-way interaction among the Model talker's productions, sound change Feature, and Age group to investigate whether age-related differences in accommodation varied across individual sound change features. The model also included random intercepts for Token and Participant, together with a by-participant random slope for the model talker's productions. In all, the output formula is:

*code_s ~ code_b + code_m * interlocutor + code_m : age : interlocutor + code_m * Feature * age + (1|token) + (1+code_m|participant)* (1)

3 Results

Table 2 outputs the fixed effects that are relevant to the research questions. Participants' baseline production significantly predicted their shadowing production. As shown in Figure 1a, speakers' percentage of target variant production is highly correlated with their shadowing performance, suggesting within-individual consistency. Neither the interaction between Model talker's productions and Interlocutor type (β = -0.085, 95%CI [-0.35, 0.49]) nor the three-way interaction among Model talker's productions, Age, and Interlocutor type (β = -0.048, 95%CI [-0.53, 0.43]) was significant, indicating that the present data did not provide statistical evidence for a difference in accommodation between human and AI interlocutors, either overall or across age groups. The corresponding confidence intervals are reported to characterize the uncertainty around these estimates. Significant interactions among Model talker's productions, sound change Feature, and Age group were found, suggesting age-related differences in accommodation varied across individual Cantonese sound change features.

Table 2. Selected GLMM output.

Predictor	β	SE	95%CI	z	p
code_b	3.153	0.202	[2.76, 3.55]	15.60	<.001***
code_m	5.332	0.945	[3.48, 7.19]	5.64	<.001***
code_m × interlocutor	-0.085	0.285	[-0.35, 0.49]	-0.30	.766
code_m × age × interlocutor	-0.048	0.245	[-0.53, 0.43]	-0.19	.846
code_m × initial kw-k × age	16.004	31.165	[-93.49, 125.51]	0.51	0.608
code_m × initial k-kw × age	5.030	1.492	[2.10, 7.96]	3.37	<.001***
code_m × syllabic ŋ-m × age	3.131	1.201	[0.77, 5.49]	2.61	.009**
code_m × coda ŋ-n × age	3.109	1.123	[0.90, 5.31]	2.77	.006**
code_m × coda n-ŋ × age	2.524	1.377	[-0.18, 5.53]	1.834	.067.
code_m × initial n-l × age	2.479	1.168	[0.19, 4.77]	2.12	.034*

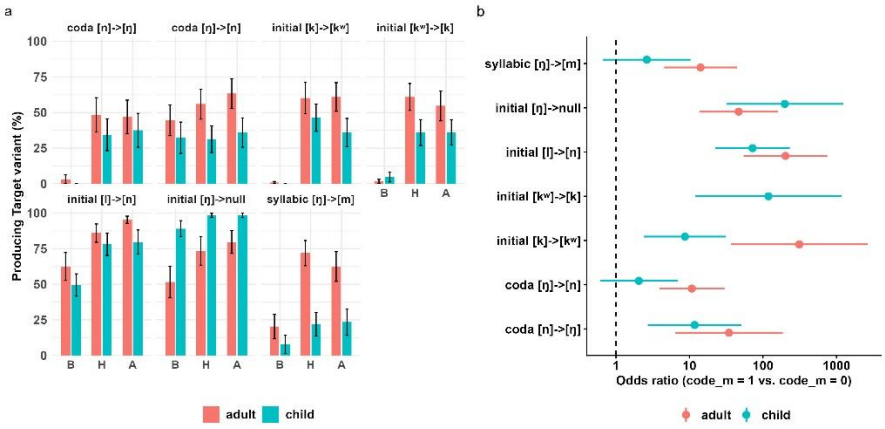


Fig. 1. (a). Mean percentage of target variant productions for the target trials (code_m = 1) under the baseline (B), human-shadowing (H), and AI-shadowing (A) conditions. Error bars indicate ±1 SE. (b). Estimated odds ratios comparing participants' production of the target variant when the model talker produced the target versus the filler variant. Log odds were converted into odds ratios. Higher ratio indicates higher probability of adopting model talkers' variants. Error bars indicate 95% CI.

Follow-up estimated marginal means analyses were conducted via 'emmeans' package [23] in R to estimate probability of adopting target variants for each feature within each age group. For each Feature-Age combination, the difference in predicted log odds of participants producing the target variant in shadowing blocks (code_s = 1) between trials in which the model talker produced the target variant (code_m = 1) and trials in which the model talker produced the filler variant (code_m = 0) was calculated and visualized (Figure 1b). The initial [kʷ]-[k] merger feature in the adult group was

excluded from visualization because no [k] productions (code_s = 1) were observed for filler trials in adult group (quasi-complete separation). As shown in Figure 1b, compared to adults, children were less likely to adopt the model talkers' variants in most features. The difference was especially pronounced for the initial [k]-[kʷ] merger feature. The only feature for which children were more likely to adopt the model talkers' variants than adults was initial [ŋ] deletion.

4 Discussion and Conclusion

The present study showed that both children and adults accommodated towards the model talkers' productions to a similar extent across human and AI interlocutors. However, accommodation differed across individual sound change features, and these feature-specific patterns varied by age group. Overall, children exhibited weaker accommodation than adults for most features.

We did not find evidence showing different extent of accommodation across human and AI interlocutors, which suggests that AI interlocutors can elicit accommodation patterns comparable to those elicited by human interlocutors. The finding aligns with previous research reporting similar alignment patterns between human and AI talkers for prosodic features in emotionally expressive utterances [24]. It could be explained by the Computers as Social Actors (CASA; [25]) account, which proposes that humans tend to treat computers as social interlocutors and apply similar social norms to interactions with computers as they do with human interlocutors. However, the interpretation that accommodation is automatic cannot be fully ruled out. Humans have been shown to exhibit some 'overlearned social behaviors', such as acting politely to computers [26], indicating that certain social behaviors towards AI agents may occur automatically without explicit social motivation, at least in the short term [27]. If accommodation towards AI agents is also driven by such automatic processes, simply copying the habitual patterns developed through HHI, the influence of AI agents on the accommodation and potential spread of sound change features may become intertwined with human-mediated processes and operate implicitly. It will be challenging to tease it apart from human influence. Nevertheless, it should be noted that the lack of evidence of different accommodation effects between human and AI interlocutors are based on our binary coding of variant adoption. It remains possible that speakers adopted the variants categorically to a similar extent across two interlocutor types, while still showing subtle differences in fine-grained phonetic realizations. Further research incorporating continuous acoustic measures is needed to explore this possibility.

The finding of generally weaker effect on children than in adults is inconsistent with previous studies on phonetic imitation of English VOT, where children showed greater imitation than adults [28] or a comparable degree of imitation [29]. This discrepancy may arise because the current study targeted at a different dimension of accommodation. In our study, accommodation involved adopting a categorically different variant, whereas accommodation in VOT studies involved gradual changes in fine-grained phonetic details. Such categorical accommodation is more likely to be affected by linguistic ability and social skills, the development of which is showed to be positively associated

with adaptive communication skills [30]. It is reasonable to expect adults who possess more developed linguistic ability and social skills to show greater accommodation than children. Future studies could combine these two dimensions of accommodation to elucidate the factors underlying generational differences in accommodation. Besides, the stimuli of the current study involved sound change features, which may be more socially salient to adults than children. Adults may therefore pay more attention to them, potentially resulting in greater accommodation. Furthermore, these features may be at different stages of change across generations, which may also have influenced the accommodation patterns observed in children and adults.

Figure 1a shows the discrepancy of adopting target variants between the young adults and children in the baseline condition for individual sound change feature, suggesting that an age gap of approximately ten years (i.e., the average age difference between the two groups) is sufficient to yield differences in the use of ongoing sound change variants. Particularly, for initial [ŋ] deletion, children already produced the target variants in most baseline tokens. Consequently, the higher proportion of target variant productions observed in the shadowing blocks largely reflects the maintenance of their existing production rather than increased accommodation. This explains why initial [ŋ] deletion was the only feature for which children appeared more likely than adults to adopt the model talkers' variants. For initial [n]-[l] merger, adults and children showed comparable accommodation effects, likely because this feature is strongly associated with speech register [31]. The more formal register elicited by the shadowing task may have increased the use of the [n] variant, potentially masking the accommodation effect. For the other features, adults showed a stronger accommodation effect than children. All these findings contributed to a complex picture of children and adults' roles in sound change. As suggested in [12], different age groups may contribute to language change through different mechanisms of linguistic modification and participation.

This study is the first to investigate cross-generational phonetic accommodation of Cantonese sound change features in the era of artificial intelligence. Children accommodated less than adults, similar between human and AI interlocutors but conditioned by progress of individual sound change features. Expanding this line of research to a wider age range, other languages, and different types of AI interlocutors will provide a more comprehensive understanding of sound change in the AI era.

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